

Numerical PDEs Homework #0

For all problems asking for a graph: you do not need to write up the method you programmed. Including plots of your solution and convergence is sufficient for full credit. The ideal homework will have one, information-dense and properly labeled figure with a caption and a *brief* discussion. **You must type up your HW.** I don't care if it's word or Tex as long as it's well organized, easy to follow, and submitted as a pdf.

Problem 1: Finite differences Write a matlab code that uses finite differences to compute the first derivative of

$$f(x) = \exp(\sin(x))$$

on the interval $0 \leq x \leq 2\pi$. Use second-order centered differences on internal grid points and first-order one-sided differences at endpoints.

1. Show a graph of the exact analytic solution $f'(x)$ and your numerical solution $f'_n(x)$.
2. Show a relative error convergence plot by plotting the relative error on the y-axis and the grid size N on the x-axis. Measure relative error as

$$Err = \frac{\|f'(x) - f'_n(x)\|_p}{\|f'(x)\|_p}.$$

Plot the error using both the $p = \infty, 2$ norms in a log-log scale. The relative errors should be lines with slopes 1 and 3/2 respectively, i.e. it scales with $\frac{1}{N}, \frac{1}{N^{3/2}}$. Extra Credit: The 3/2 slope is strange, why is that happening?

Problem 2: Second-order derivatives Repeat problem 1 but compute the second derivative $f''(x)$. Assume periodicity and wrap around the grid instead of using one-sided finite differences.

Problem 3: ODE Consider the ODE

$$\frac{d^2u}{dx^2} + \sin(x) \frac{du}{dx} + u(x) = f(x)$$

defined on $0 \leq x \leq 5$. Solve the ODE using second-order finite difference methods with the Dirichlet conditions. Use $u(x) = \sin(x)$ to test your solution by finding boundary conditions and the necessary forcing function $f(x)$. Show an error plot that proves your method converges with second order.

Problem 4: Poisson Equation Use finite difference methods to solve the Poisson equation $\nabla^2 u = f(x, y)$ in the domain $x \in [0, 5], y \in [0, 5]$. Use the test solution $u(x, y) = \sin(x) \cos(y)$.

1. Apply Dirichlet conditions on all boundaries so that $u(x, y)$ satisfies the PDE and the boundary conditions, and show a plot of your solution and a second order convergence plot (similar to the first three problems).
2. Modify your code to apply Neumann conditions on all boundaries. **This won't work.** Why not? – both in terms of the linear algebra and in terms of the actual PDE we want to solve. Suggest a solution to the issue.

$$1) \text{ backward: } \frac{\partial f}{\partial x} = \frac{f(x) - f(x-h)}{h}$$

$$\text{Central: } \frac{\partial f}{\partial x} = \frac{f(x+h) - f(x-h)}{2h}$$

$$\text{forward: } \frac{\partial f}{\partial x} = \frac{f(x+h) - f(x)}{h}$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{u(x+h) - 2u(x) + u(x-h)}{h^2}$$

$$f = e^{\sin(x)} \\ \frac{\partial f}{\partial x} = \cos(x) e^{\sin(x)}$$

(Gemma)
What I used AI for:

- 1) formatting graphs syntax
- 2) I thought 1st order would be better so I loaded code and asked how I was calculating wrong bit things or I was okay
- 3) When given AI find error (missing gbs) in p-oo case

Problem 3: ODE Consider the ODE

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$$-\cancel{\sin(x)} + \sin(x) \cos(x) + \cancel{\sin(x)} = f(x)$$

$$f(x) = \sin(x) \cos(x)$$

BCs:

$$\sin(0) = 0$$

$$\sin(5) \approx -0.95892$$

*Solving f, BCs
converges w/
second order

$$\phi''(x) = \frac{\phi(x-\Delta x) - 2\phi(x) + \phi(x+\Delta x)}{\Delta x^2}$$

$$\frac{\phi(x+\Delta x) - \phi(x-\Delta x)}{2\Delta x} \approx \phi'(x)$$

$$\frac{d^2 u}{dx^2} + \sin(x) \frac{du}{dx} + u(x) = f(x)$$

$$\frac{u_{i+1} - 2u_i + u_{i-1}}{h^2} + \sin(x) \frac{u_{i+1} - u_{i-1}}{2h} + u_i = \sin(x) \cos(x)$$

$$u_{i-1} \left(\frac{1}{h^2} - \frac{\sin(x)}{2h} \right) + u_i \left(1 - \frac{2}{h^2} \right) + u_{i+1} \left(\frac{1}{h^2} + \frac{\sin(x)}{2h} \right) = \sin(x) \cos(x)$$

$$\begin{bmatrix} 1 \\ \frac{1}{h^2} - \frac{\sin(x)}{2h} \\ 1 - \frac{2}{h^2} \\ \frac{1}{h^2} + \frac{\sin(x)}{2h} \end{bmatrix}$$

$$\cos(s) \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ \vdots \\ u_{n-1} \\ u_n \end{bmatrix} = \begin{bmatrix} \sin(x_0) \cos(x_0) \\ \vdots \\ \sin(s) \cos(s) \end{bmatrix}$$

$\uparrow$
 forces u_1 to $\sin(s)$

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$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f(x, y)$$

$$-\sin(x)\cos(y) - \sin(x)\cos(y) = f(x, y)$$

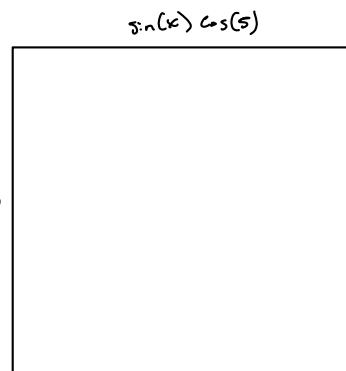
$$\frac{u_{i-1}^j - 2u_i^j + u_{i+1}^j}{h^2} + \frac{u_i^{j-1} - 2u_i^j + u_i^{j+1}}{h^2} = -2\sin(x)\cos(y)$$

$$f(x, y) = -2\sin(x)\cos(y)$$

$$\frac{u_{i-1}^j + u_{i+1}^j + u_i^{j-1} + u_i^{j+1} - 4u_i^j}{h^2} = -2\sin(x)\cos(y)$$

$$u_{i-1}^j + u_{i+1}^j + u_i^{j-1} + u_i^{j+1} - 4u_i^j = -2h^2\sin(x)\cos(y)$$

$$\begin{bmatrix} 0 & 0 & h^2 \cos(y) & 0 & 0 \end{bmatrix} \begin{bmatrix} u_i^j \end{bmatrix} = \begin{bmatrix} -2h^2 \sin(x) \cos(y) \end{bmatrix}$$



1) Why is solution not-unique

2) How do I fix it

you can fix it anywhere

fix to

now is anywhere and still satisfy PDE & BCs